

RONIE MARTÍNEZ

AI, ML & DL | DataScience | Development
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ABOUT ME

I am a dedicated and versatile professional with a robust background in Artificial Intelligence, Machine Learning, and Data Science, complemented by a solid foundation in software development. My academic journey, crowned with a Master's degree in Artificial Intelligence and a Bachelor's in Electronics and Telecommunications, has provided me with a deep understanding of both theoretical frameworks and their practical applications. I thrive on tackling complex challenges and have a proven track record of developing innovative solutions, ranging from sophisticated deep learning models for medical diagnostics to dynamic and interactive web applications.

My professional experience is diverse, spanning freelance consulting, research advising, and engineering roles. I have a passion for education and have been actively involved in teaching and mentoring, sharing my knowledge in areas like programming, data science, and AI. As a lifelong learner, I am constantly exploring new technologies and methodologies to stay at the cutting edge of the industry. I am proficient in a variety of programming languages and tools, including Python, JavaScript, Java, C++, MATLAB, and the full web stack. My goal is to leverage my skills and experience to contribute to meaningful projects that push the boundaries of technology and create a positive impact.

PROFESSIONAL EXPERIENCE

Freelancing @ Research UIO

Jan 2013 - Current

- Tutoring, private classes, consulting, project development, and research advising in artificial intelligence, machine learning, data science, digital communications, algebra, and basic robotics.
- License plate detection and recognition systems using WPOD-NET, the MobileNetV2 model, and image processing.
- Removal of the pectoral muscle and segmentation for an aided breast cancer diagnosis system using image processing.
- Valvulopathies detection system from ECG using 1-D CNNs.
- Machine Learning-Based Classification for Cocoa Beans through images.
- Predictive Modeling and Machine Learning for Private Logistics.
- Data visualization for Covid-related information.
- Machine Learning-Based Prediction for Electric Transformer's Load Capacity.

Administrative and Projects Assistant | Dismafer

Oct 2015 - Nov 2021

- Accounting, administrative and manager assistant.
- Project consulting for space automation within smart homes.

- Training for the suitable use of smart home device for smooth automation.

Junior Engineer | Linkzartec

Aug 2014 - Jan 2015

- Testing implementation and evaluation of the electronics systems at "Calderon Teaching Hospital" i.e. audio, illumination, access, CCTV, fire control, queuing systems and intercoms.
- Electronic systems design for Telefonica Ecuador CAVs (customer service and sales centers).
- Setting and maintenance of DVRs (digital video recorders) and MDVRs (mobile digital video recorders).

Instructor/Exhibitor | Alamos High School

Mar 2014 - Apr 2014

- Conducted workshops in Python programming, HTML and CSS.
- Development of a vehicle controlled by Wi-Fi and live streaming using Raspberry Pi.

TEACHING AND LABORATORY

Lecturer

2025 - current

Information and Computer Science Department DICC, Escuela Politécnica Nacional EPN

- Programming I with C++ - BEng. in Computer Science & BEng. in Data Science and Machine Learning.
- Programming II with Java - BEng. in Data Science and Machine Learning.
- Computer Architecture - BEng. in Computer Science, BEng. in Software & BEng. in Data Science and Machine Learning.

Lecturer

2023 - 2025

Computer Science School, Universidad Internacional del Ecuador UIDE

- Data Storage and Attachment - Master's Degree in Information Systems and Business Intelligence.
- Data Analysis, Modelling and Management in Data Science - Master's Degree in Data Science and Machine Learning.
- Machine Learning - Master's Degree in Information Systems and Business Intelligence.
- Image and Signal Processing - Master's Degree in Applied Artificial Intelligence (Lecturer Assistant).
- Technology and Society - BS. in Information Technology.
- Programming Logic - BS. in Information Technology.
- Introduction to Internet Networking - BS. in Information Technology.
- Info Model, Store & Retrieval - BS. in Information Technology.
- Introduction to Java Technologies - BS. in Information Technology.
- Applied Programming Language for Information Technology - BS. in Information Technology.
- Web Programming for Human-Computer Interaction - BS. in Information

Technology.

- Data Analysis, Modelling, and Management in Data Science - Master's Degree in Data Science and Machine Learning.
- Web Programming for Human-Computer Interaction - BS in Information Technology.
- Intermediate Database Management Systems - BS in Information Technology.

Invited Lecturer

2024

Universidad Iberoamericana del Ecuador - UNIBE

Teaching Assistant

2013 - 2014

Department of Electrics and Electronics, Universidad de las Fuerzas Armadas - ESPE

- Supervised and reviewed a course named "Data Communication" throughout the provision of feedback and the execution of mentoring sessions to undergraduate students.
- Reviewed and marked exams based on results and performance.
- Taught a module about error detection and correction over communication channels.

Laboratory Assistant

2013

Department of Electrics and Electronics, Digital Television Laboratory, Universidad de las Fuerzas Armadas - ESPE

- Prepared test beds for CONATEL to validate TVs' performance under the ISDB-Tb standard for DTT (Digital Terrestrial Television).
- Used the tools from the Digital Television Laboratory to satisfy and comply the tests proposed by MINTEL for the corresponding approval and television certification.
- Used software (VillageFlow, StreamXpress y StreamXpert) and hardware (transmission and reception antennas, radio frequency amplifiers, televisions, decoders, etc.) relevant to perform certification of receiving equipment
- Developed and tested interactive Ginga applications for DTT using Eclipse Galileo y Ginga Windows.

RESEARCH EXPERIENCE

Peer Reviewer

2025 - current

- Latin-American Journal of Computing (LAJC)

Co-Investigator

2024 - current

Biomedical Data Science Laboratory (BDS_Lab), Universidad Internacional del Ecuador UIDE

- Metabolic Pathways of Alternative Lengthening of Telomeres in Pan-Carcinoma.
- Exploration of Andinan and Galapagos pytho-metabolites as potential

inhibitors for colorectal adenocarcinoma and breast cancer: a docking and molecular dynamics analysis.

- Mindfulness-based intervention model with real-time respiratory flow monitoring for enhancing cognitive function and well-being.

Master's Research

2019 - 2020

Department of Engineering and Technology, Universidad Internacional de La Rioja UNIR

- Conducted a comparison of deep learning models used for melanoma skin cancer detection, selected the best option to attempt the reduction of the related mortality rate through early detection, and proposed a model for a Computer Aided Diagnosis (CAD) System.
- Conducted critical discussions according to the results of the models' performance based on specific metrics and the trend of using convolutional neural networks (CNN), data augmentation and transfer learning.

Research Assistant

2013 - 2017

WiCOM Energy Research Group, Universidad de las Fuerzas Armadas - ESPE

- Evaluated the performance for digital television transmitter stations using ISDB-T standard over AWGN and fading channel.
- Modeled a complete DTT (ISDB-T) system following a mathematical approach for each process and developed a simulation environment to support and compare theoretical and simulated results.
- Proposed a Cloud-Tx model to enhance the performance in limited conditions (e.g., Low SNR scenarios), with excellent results regarding BER and signal quality over all the three definition modes in the ISDB-T standard.

Research Assistant

2013 - 2017

Department of Electrics and Electronics, Universidad de las Fuerzas Armadas - ESPE

- Designed and developed a tool oriented to enhance fine motor skills in people with Down Syndrome.

Research Assistant

2013 - 2017

Computer Networks Research Group, Universidad de las Fuerzas Armadas - ESPE

- Co-created the novel research group with the cooperation and mentoring of an international researcher from Brazil.
- Conducted literature review and analysed neighbors selection algorithms regarding on-demand video services for BitTorrent-based systems.
- Assisted and proposed mechanisms for distribution of multimedia content over computer networks.

PROJECTS

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- **License Plate Detection and Recognition:** Developed a system for license plate detection and

recognition using WPOD-NET and MobileNetV2.

- **Breast Cancer Diagnosis Aid:** Created a system for the removal of the pectoral muscle and segmentation for an aided breast cancer diagnosis system using image processing.
- **Valvulopathies Detection:** Built a valvulopathies detection system from ECG using 1-D CNNs.
- **Predictive Modeling for Logistics:** Designed and implemented predictive models and machine learning algorithms for private logistics.

SKILLS

Artificial Intelligence, Machine Learning, Data Science, Web Development, Python, C++, Java, JavaScript, HTML, CSS, React, Node.js, SQL, Git, MATLAB, Deep Learning, Generative Adversarial Networks

EDUCATION

MS in Artificial Intelligence

Universidad Internacional de La Rioja UNIR, La Rioja, Spain - 2020

BSc in Electronics and Telecommunications

Universidad de las Fuerzas Armadas - ESPE, Quito, Ecuador - 2017

PUBLICATIONS (SELECTED)

- Herrera-Granda, E. P., Cabascango-Flores, P. V., Sandoval-Palis, I. P., Sánchez-Almeida, T., Villota-Cadena, Á. P., Aza-Espinosa, M. J., Martínez, R., & Herrera-Granda, D. E. (2026). Integrating Three-Parameter Logistic IRT Models and Confirmatory Factor Analysis for Multidimensional Assessment of Academic Performance and Associated Factors in University Leveling Programs. *Applied Sciences*, 16(12), 6248.
- Martínez, R. (2025). Ethical Governance of Neurotechnologies in Latin America. *NeuroData*.
- Martínez Gordon, R. S., López Mendoza, X. P., Freire Díaz, J. P., & Cózar Cisneros, A. E. (2025). Enseñanza Inteligente de la Matemática: Aplicaciones del Machine Learning y la IA en la Educación Superior. *Páginas Brillantes Ecuador*. Quito, Ecuador.
- Mardones, A. M., Vega, R. G. A., Trujillo, F. K., Martínez, R., Acarapi, J. M., & Bastidas, J. O. G. (2025). Gamification and digital environments in higher education: Enhancing motivation and experiential learning. In *Seminars in Medical Writing and Education* (Vol. 4, p. 5). AG Editor (Argentina).
- Martínez Gordon, R. S., Becerra Zurita, M. A., Guallichico Guallichico, L. A., & Paguay Salguero, J. F. (2025). Matemáticas aplicadas a proyectos del mundo real: Resolución de problemas con TIC y pensamiento crítico. *Páginas Brillantes Ecuador*. Quito, Ecuador.
- Manzano, J. E. E., Martínez, R., Armijos, F. A. P., García, M. M. G. (2025). Evaluación del uso del enfoque STEAM en el rendimiento académico en

- secundaria. Sinergia Académica, 8(5), 845-856.
- Naranjo, B. M. M., García, M. M. G., Manzano, J. E. E., Martínez, R. (2025). Uso de simuladores virtuales en la enseñanza de conceptos abstractos. Sinergia Académica, 8(4), 713-727.
 - Arango, M. D. L. V., Martínez, R., Castillo, E. A. R., Mendoza, R. R. P. (2025). Desarrollo de la comunicación efectiva, el trabajo colaborativo y la capacidad para la resolución de problemas, a través del Aprendizaje Basado en Proyectos.
 - Reyes, A. T. C., Pallasco, L. R. A., Alexandra, P. G. Y., Quishpe, E. R. A., Martínez, R. (2025). Integración de metodologías de aprendizaje basado en la investigación para fomentar competencias científicas en estudiantes universitarios.
 - Armendáriz-Castillo, I., García-Cárdenas, J., Espinosa, P., Hidalgo-Fernández, K., Peña-Zúñiga, L., Martínez, R., et al. (2025). Metabolic pathways of Alternative Lengthening of Telomeres in pan-carcinoma. PloS one, 20(2), e0314012.
 - Martínez, R., Olmedo, G., & Paredes, N. (2017, August). Analytical performance of digital terrestrial television transmitter stations for mobile broadcasting using ISDB-T standard over AWGN and fading channel. In 2017 IEEE Colombian Conference on Communications and Computing (COLCOM).

COURSES & CERTIFICATIONS

📄 Diploma in Design and Management of Educational Projects (65H) (2026)

- Project-based learning in schools in rural contexts - Universidad Austral (22H)
- Designing projects to promote well-being - Universidad de los Andes (14H)
- Introduction to project management - Tecnológico de Monterrey (25H)
- Path to Excellence in Project Management - Pontificia Universidad Católica de Chile (7H)
- Project risk and change management - University of California, Irvine (5H)

📄 Diploma in SDGs and Sustainable Education (52H) (2025)

- Through purple glasses: gender equality in schools - Universidad Nacional Autónoma de México (8H)
- Introduction to sustainability: towards a possible future - Universidad Nacional Autónoma de México (11H)
- SDGs

and sustainability: a view from schools - Universidad de los Andes (17H)

- Introduction to University Social Responsibility - Pontificia Universidad Católica del Perú (2H)
- Sustainable Development Goals for Latin America and the Caribbean - Universidad de los Andes (14H)

📄 **Diploma in Design and Educative Gamification (58H) (2025)**

- Designing Learning Innovation - Politecnico di Milano (24H)
- e-Learning Ecologies: Innovative Approaches to Teaching and Learning for the Digital Age - University of Illinois Urbana-Champaign (19H)
- Teaching with active pedagogy mediated by digital technology - Universidad de los Andes (26H)

📄 **Microsoft AI & ML Engineering Professional Certificate (In progress)**

- Foundations of AI and Machine Learning
- AI and Machine Learning Algorithms and Techniques
- Building Intelligent Troubleshooting Agents
- Microsoft Azure for AI and Machine Learning
- Advanced AI and Machine Learning Techniques and Capstone

📄 **Meta Front-End Developer Professional Certificate (In progress)**

- Introduction to Front-End Development
- Programming with JavaScript
- Version Control
- HTML and CSS in depth
- React Basics
- Advanced React
- Principles of UX/UI Design
- Front-End Developer Capstone
- Coding Interview Preparation

📄 **Amazon Junior Software Developer Professional Certificate (In progress)**

- Introduction to Software Development
- Programming with Java
- Data Structures and Algorithms
- Database Management with Java and SQL
- Full Stack Web Development
- Generative AI in Software Development
- Application Development

📄 **Diploma in Inclusive Education and**

Disability (49H) (2025)

- Designing Learning Innovation - Politecnico di Milano (24H)
- Diversity and Inclusion in Education - University of Glasgow (17H)
- Understanding disability in university settings - Universidad Nacional Autónoma de México (4H)
- Disability Inclusion in Education: Building Systems of Support - University of Cape Town (14H)
- Social inclusion of people with disabilities from a rights-based perspective - Universidad de Chile (14H)

☒ Diploma in Artificial Intelligence

Applied for Teaching (55H) (2024)

- Artificial Intelligence (AI): Interactions and Prompts - Universidad de Palermo (14H)
- AI for Educators: Transform Your Teaching with ChatGPT - Universidades Anáhuac (9H)
- ChatGPT - Uses and Strategies - Tecnológico de Monterrey (18H)
- Generative AI in the Classroom - Universidad Nacional Autónoma de México (11H)

☒ Diploma in 21st Century Practical

Teaching Skills (40H) (2024)

- Practice-based teacher training to develop 21st century skills - Pontificia Universidad Católica de Chile (27H)
- What future for education? - University of London (19H)

☒ Introduction to NoSQL Databases (18H) (2024)

☒ Introduction to MongoDB (15H) (2024)

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Inclusive educational attention strategies (2024)

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Online Teaching Starter Pack (2024)

☒ Practical Data Science with MATLAB Specialization - MathWorks (67H) (2021)

- Exploratory Data Analysis with MATLAB (16H)

- Data Processing and Feature Engineering with MATLAB (20H)
- Predictive Modeling and Machine Learning with MATLAB (18H)
- Data Science Project: MATLAB for the Real World (13H)

☒ **Generative Adversarial Networks**

(GANs) Specialization - DeepLearning.AI (82H) (2020)

- Build basic Generative Adversarial Networks (GANs) (29H)
- Build better Generative Adversarial Networks (GANs) (28H)
- Apply Generative Adversarial Networks (GANs) (25H)

☒ **MATLAB Programming for Engineers and**

Scientists Specialization - Vanderbilt University (113H) (2020)

- Introduction to Programming with MATLAB (35H)
- Mastering Programming with MATLAB (55H)
- Introduction to Data, Signal, and Image Analysis with MATLAB (23H)

☒ **Deep Learning Specialization. Master**

Deep Learning, and Break into AI - DeepLearning.AI (125H) (2020)

- Neural Networks and Deep Learning (24H)
- Improving Deep Neural Networks (23H)
- Structuring Machine Learning Projects (6H)
- Convolutional Neural Networks (35H)
- Sequence Models (37H)